

APPM 6470 Project Description

Spring 2026

Overview

The course project is an opportunity to explore partial differential equations beyond homework-style problems and to engage more deeply with theory, applications, or research directions related to the course. The goal is to develop independence in reading, analyzing, and communicating PDE ideas. Think of this as a small step toward research-level mathematical work. Your project should focus on **one** of the following directions:

Option A: Deep Dive into a Course Topic

Study a topic from class in greater depth. This could include:

- One or multiple theorems whose proofs were only sketched or skipped.
- A method we introduced, but did not fully develop
- A class of PDEs related to those studied in the class
- Numerical, analytical, or qualitative aspects of a model PDE

You should go beyond the lecture notes and consult outside sources.

Option B: Application to Your Research

Apply PDE tools from this course to a problem in your own research area. Examples include

- Modeling a phenomenon with a PDE framework
- Analyzing an existing PDE model relevant to your field
- Exploring well-posedness, stability, or asymptotics
- Connecting PDE theory to data, numerics, or modeling assumptions

The emphasis is on demonstrating how course tools transfer to real problems.

Option C: New Research Directions

Explore a new or emerging research direction that uses PDE methods. This might include:

- PDEs in biology, social systems, climate science, or materials
- Kinetic or multiscale limits
- Nonlocal PDEs
- PDEs on networks or manifolds
- Data-driven or learning-based PDE models

This option encourages creative exploration.

Option D: Paper Reading and Exposition

Select a research or survey paper related to PDEs and write a clear exposition of it. Your goal is to make the main ideas accessible to classmates who have taken this course. Your exposition should include:

- Background and motivation
- Explanation of key results and methods
- Discussion of the main ideas of proofs (at an appropriate level)
- Context within the broader PDE literature

You are encouraged to include examples, diagrams, or simplified cases that aid understanding. The focus should be on clarity and insight rather than reproducing technical proofs in full detail.

Deliverables

Project Proposal: Due Friday, March 27, 2026

Submit a short proposal (1–2 pages) including:

1. Title and topic area
2. Motivation: why the topic is interesting or important
3. Goals: what you want to learn or demonstrate
4. Preliminary plan: key questions or methods
5. References: at least 3 initial sources

Project Report: Due Friday, April 24, 2026

Your report should include:

- Clear mathematical exposition
- Background and context
- Key results or insights
- Proper references
- Your own understanding and synthesis

Possible formats include: Expository paper (typically 8–15 pages), Modeling or analysis report, Theory-focused write-up, and an Optional computational component.

Project Presentation: To be held on the last days of class and during our final exam

You will present your work at the end of class, instead of a final exam. This will be a 10-15-minute presentation. Projects will be evaluated on: Mathematical clarity and correctness, depth of engagement, quality of explanation, evidence of independent learning, and organization and presentation. You will be expected to be present and engaged for our classmates' presentations. This will be taken into account for the grades.